

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, Java, C, C++, SQL, Swift, Bash, HTML, CSS
Frameworks & Libraries: React, Node.js, Express, SwiftUI, Bootstrap, Tailwind, LangChain
AWS & Cloud: Lambda, RDS, Aurora, S3, Cognito, Textract, API Gateway, CloudWatch, Bedrock, SQS, Glue, Amplify
AI/ML: Amazon Bedrock, Llama 3, Titan Embeddings, RAG, Vector Databases (pgvector)
Tools & Technologies: Git, Linux, Docker, PostgreSQL, MongoDB, Oracle SQL, REST APIs, WebSocket, OpenAPI

TECHNICAL EXPERIENCE

Full Stack Developer - AWS Cloud Innovation Centre, UBC Sept 2025 – Present

- Legal Aid Platform:** Enhanced application security by implementing frontend input validation with file size and character limits, upgrading PostgreSQL RDS, and migrating Lambda functions to Node.js 22 runtime, reducing security vulnerabilities by 85%.
- Resolved critical Security Hub findings by enabling comprehensive CloudWatch logging, adding API Gateway deletion protection, and sanitizing user inputs to prevent injection attacks, improving security posture by 70%.
- OER-AI Platform:** Architected full-stack RAG-based generative AI educational system, designing 50+ RESTful API endpoints with OpenAPI specifications and implementing Lambda handlers for WebSocket streaming support.
- Designed PostgreSQL database schema with pgvector extension across 25+ tables and automated migration scripts, and developed JWT token-based authentication system to ensure user anonymity while maintaining session management.
- Built Dockerized Lambda function integrating Amazon Bedrock (Llama 3 70B) to generate personalized practice materials (MCQs, flashcards, short-answer questions with AI grading) using RAG-based multimodal embeddings, featuring automated grading with constructive feedback and multi-format export (H5P, JSON, PDF), reducing content creation time by 90%.
- Presented OER-AI at AWS re:Invent 2025 watchparty at Amazon Vancouver and mentored developers at Cloud Innovation Centre hackathon.

Project Manager & Developer - Coast Capital Savings Jan 2025 – May 2025

- Led a team of 8 developers to build an OCR-based document processing system for mortgage applications, loans, and new account openings as part of UBC's CPSC 319 Software Engineering Project Course, processing 3 document types with 95% accuracy.
- Designed and implemented a data extraction pipeline to streamline document handling, reducing manual processing time by 75% and error rates by 60%.
- Used AWS Lambda to arrange serverless processing tasks such as file uploads, image conversions, document parsing, and PDF generation—triggered automatically via S3 events, handling up to 1000 documents per hour.
- Integrated AWS Textract to extract structured data from scanned images and PDFs; stored parsed data using Amazon RDS and Aurora with 99.9% uptime, and implemented Cognito authentication for secure access to document workflows for 50+ concurrent users.

PROJECTS

UBC Classroom and Course Finder Sept. 2024 – Nov. 2024
Full-stack Web Application React, TypeScript, NodeJS, Maps API

- Developed a web platform to provide students with searchable information on over 100 UBC courses and 300+ room assignments.
- Designed and implemented a RESTful API with 15+ endpoints to process and serve detailed course and room data with sub-second response times.
- Integrated Google Maps API for precise campus location mapping and applied unit and integration testing with 200+ test cases, achieving 90% code coverage.

Voice Buddy Nov. 2024
HackCamp 2024 - Best Hack for Social Good React, JavaScript, NodeJS, Python, Whisper API

- Created a web application for speech impairment practice with visual guides, integrating OpenAI's Whisper API (95% accuracy) for transcript extraction and millisecond-precision word highlighting.

Meet.Point Jan. 2025
NwHacks 2025 Winner - Stream.Place Prize React, JavaScript, GeoJSON

- Built a gamified event check-in platform with location-based validation using GeoJSON and TurfJS (10-meter accuracy), supporting 100+ concurrent users.

Recipe Finder App February 2025 - April 2025
Full-stack recipe platform JavaScript, Node.js, Oracle SQL, Express

- Built a recipe discovery app with CRUD operations managing 50+ recipes across a normalized Oracle SQL schema with 8 tables.
- Developed 20+ RESTful APIs and a dynamic vanilla JavaScript front-end with data visualization, reducing page load time by 40%.

EDUCATION

University of British Columbia Expected graduation: May. 2027
B.Sc. in Computer Science - Deans's List 2021/2022
Relevant Courses: Machine Architecture, Data Structures and Algorithms, Introduction to AI, Software Engineering, Relational Databases, Networks, Machine Learning.